Subway Turnstile Controller Report

Name: 沈博宇

Student ID: 2024531115

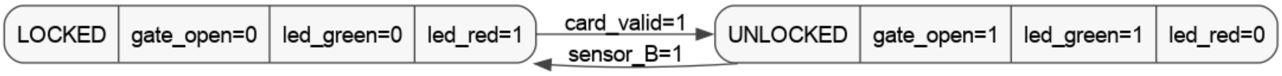
This report documents the Moore FSM design, state diagrams, simulation waveforms, and verification results for all four parts of the subway turnstile controller project. All Verilog modules were simulated with Icarus Verilog 11.0.

Design notes: asynchronous active-low reset (rst_n) in every module; sensor_B > card_valid priority in IDLE (Part 3/4); parameterized ALARM_DURATION and TIMEOUT_DURATION values.

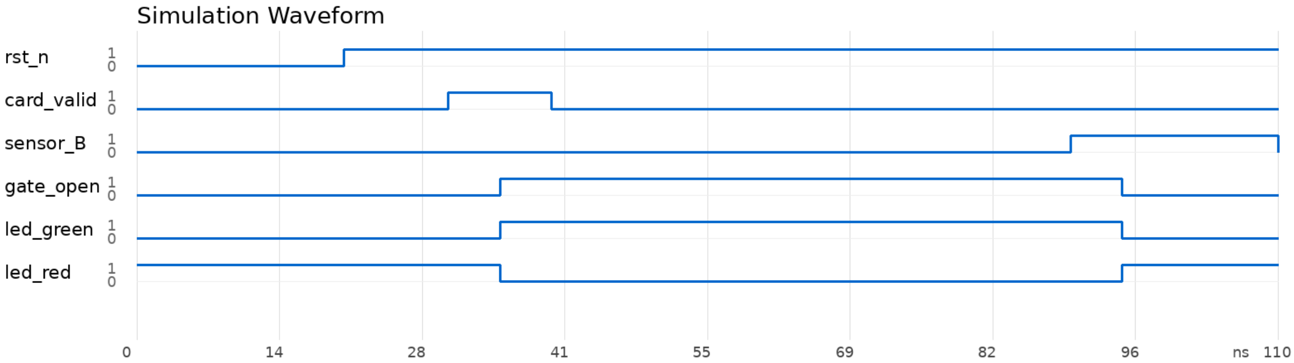
Part 1: Two-State Turnstile

A two-state Moore FSM is used. LOCKED keeps the gate closed with the red LED on. A valid card moves the controller to UNLOCKED, which opens the gate and enables passage. sensor_B closes the gate after the passenger exits.

State Diagram



Simulation Waveform



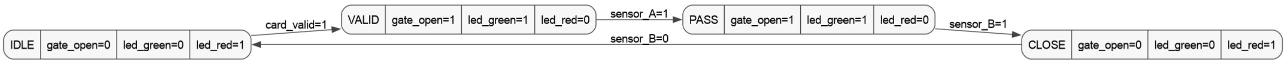
Verification

Compile: PASS | Simulate: PASS | card_valid opens gate; sensor_B closes gate. Reset holds LOCKED.

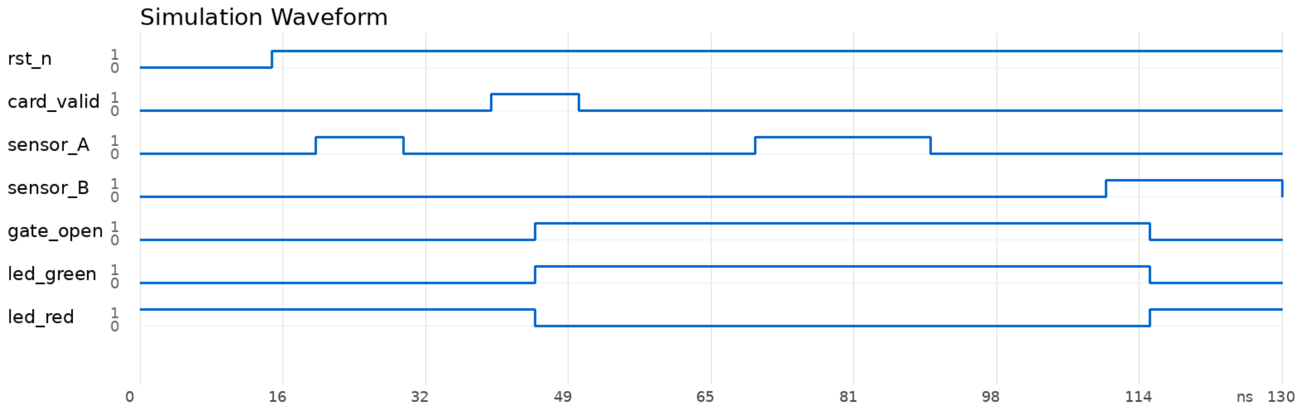
Part 2: Basic Access Control

The controller is extended to a four-state Moore FSM. IDLE waits for a card, VALID waits for lane entry at sensor_A, PASS waits for exit at sensor_B, and CLOSE holds the gate shut until sensor_B is released.

State Diagram



Simulation Waveform



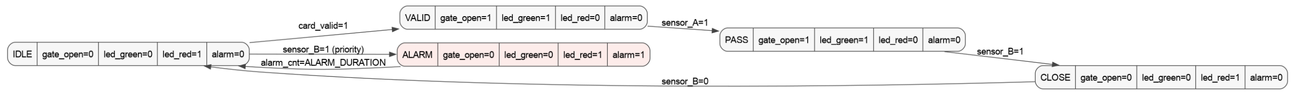
Verification

Compile: PASS | Simulate: PASS | sensor_A ignored in IDLE. Full forward chain: card → sensor_A → sensor_B → close → IDLE. Student testbench confirms both edge cases.

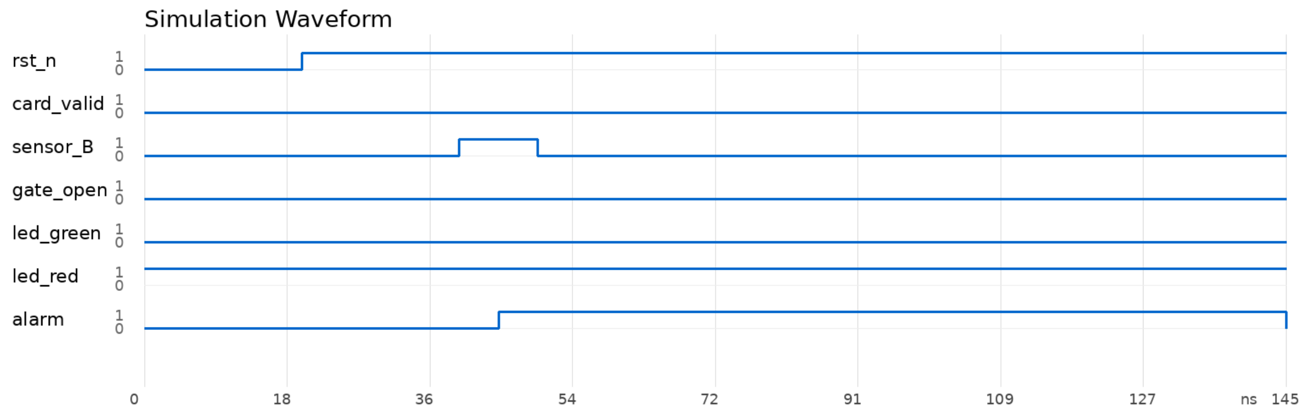
Part 3: Reverse Intrusion Detection

Part 3 adds an ALARM state and an alarm counter. In IDLE, sensor_B has higher priority than card_valid, so reverse intrusion is detected before any card action. The alarm stays high for ALARM_DURATION clock cycles, while the gate remains closed and the red LED remains on.

State Diagram



Simulation Waveform



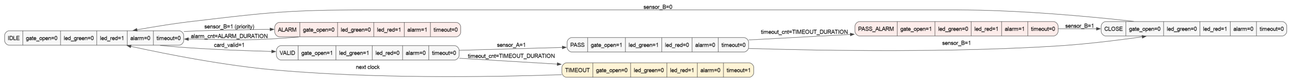
Verification

Compile: PASS | Simulate: PASS | sensor_B in IDLE triggers ALARM (priority over card_valid). Alarm duration = 10 clock cycles (ALARM_DURATION). Gate closed throughout.

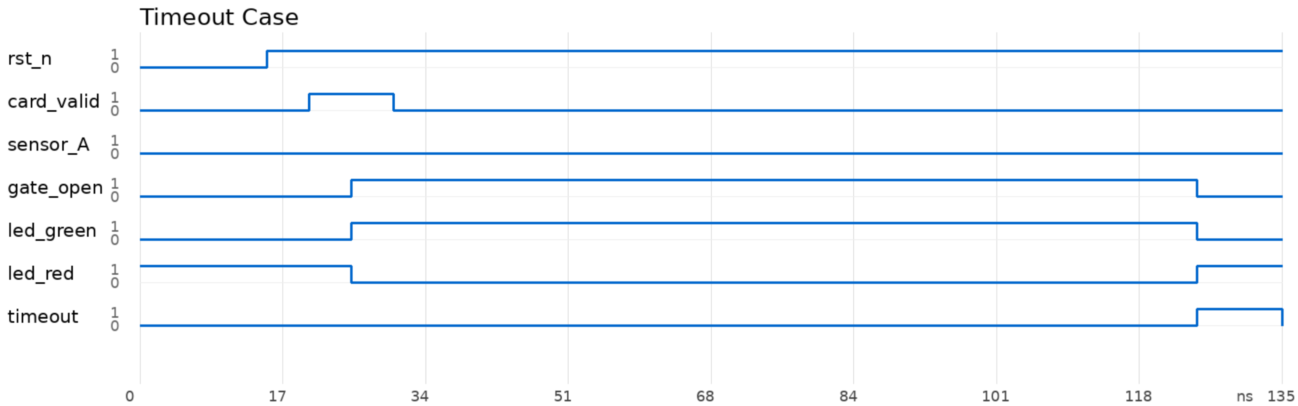
Part 4: Timeout Auto-Close

Part 4 keeps the reverse-intrusion alarm and introduces two more behaviors. In VALID, a timeout counter auto-closes the gate and asserts timeout when no passenger enters before TIMEOUT_DURATION. In PASS, the same timeout concept raises PASS_ALARM if a passenger stays in the lane too long. PASS_ALARM keeps the gate open, drives alarm high, and waits for sensor_B before finishing through CLOSE.

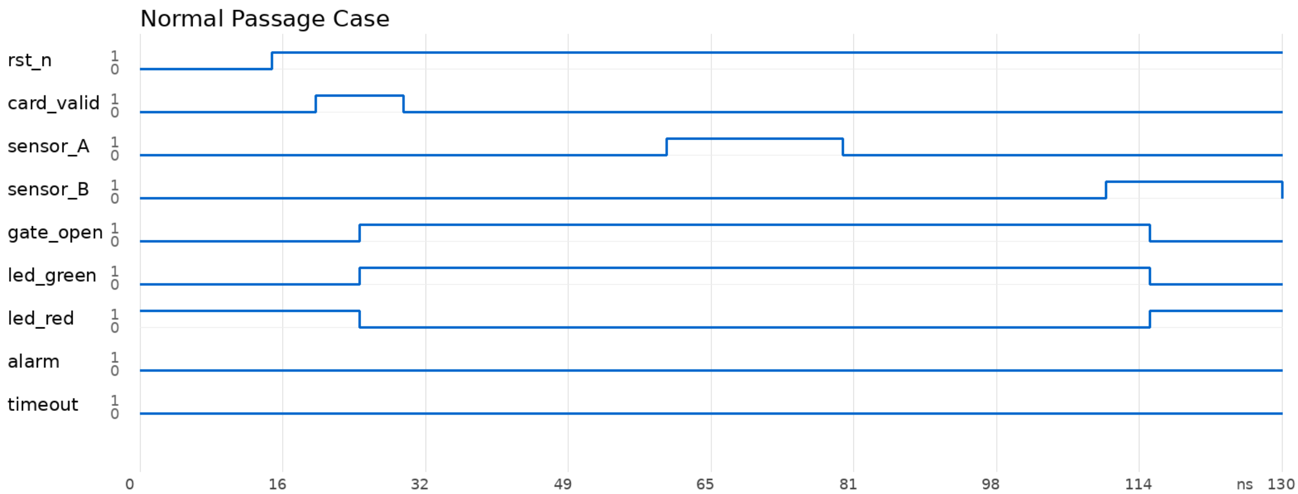
State Diagram



Timeout Case



Normal Passage Case



Verification

Compile: PASS | Simulate: PASS | Timeout case (4a): card → no sensor_A for 10 cycles → TIMEOUT. Normal case (4b): card → sensor_A → sensor_B, no alarm, no timeout. Both student testbenches verify correct behavior.

Conclusion

All four parts of the subway turnstile controller design have been compiled and simulated successfully using Icarus Verilog 11.0. The design progressively builds a feature-complete Moore FSM controller with 2, 4, 5, and 7 states respectively, covering turnstile access control, reverse intrusion detection, and timeout auto-close.

Part	Module	Testbench	Compile	Simulate
Part 1	part1.v	part1_tb.v (provided)	PASS	PASS
Part 2	part2.v	part2_tb.v (written)	PASS	PASS
Part 3	part3.v	part3_tb.v (provided)	PASS	PASS
Part 4	part4.v	part4a_tb.v / part4b_tb.v (written)	PASS	PASS

Key Design Features

• Consistent Moore FSM architecture across all parts

• Asynchronous active-low reset (rst_n) in every module

• Parameterized delay values: ALARM_DURATION, TIMEOUT_DURATION

• Priority-based signal handling: sensor_B > card_valid in IDLE

• Progressive state expansion: 2 → 4 → 5 → 7 states

• Student-written testbenches for Parts 2 and 4 verify key scenarios

• Auto-generated report with Graphviz FSM diagrams and VCD waveforms

Submission checklist: 4 Verilog modules, 3 student-written testbenches (part2_tb.v, part4a_tb.v, part4b_tb.v), report.pdf. All deliverables match the project requirements.